# Path-Robust Risk Controls for Intraday Mean Reversion in NSE Equities

**M6 Draft Project | MScFE 690 Capstone | Student Group 17838**

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**Abstract**

*Stop and profit levels can limit individual losses and close the trade before a recovery occurs. Do they enhance net returns or reduce downside risk for a fixed intraday mean-reversion strategy in five stocks of the National Stock Exchange of India? We match the time-only, stop-only or combined exits in terms of entries, position size and expiry, including stated costs and delays and two possible minute-price paths. There are 443 scheduled sessions in the development sample, of which 360 or 381 are complete-case sessions, under the two timestamp interpretations, respectively. Validation has 148 complete sessions under both interpretations. Combined-minus-time mean returns are negative in both stages (-0.2325 and -0.1273 basis points per day in development, and -0.6885 and -0.3855 in validation). Time-only gross returns are positive but less than what trading costs were assumed to be. Stop-based exits have lower observed expected shortfall but higher funded drawdown. Development does not have an ambiguous reference episode; validation has an ambiguous reference episode in one of 260 end-labelled episodes, and no reversal in the return ranking under the other price path. These sparse observations do not provide a general indication of path robustness. The intervals are exploratory since bootstrap diagnostics indicate undercoverage, including an independent simulated case. A reproducible conditional comparison of exit rules and explicit execution, accounting and inference limits is provided. The final test remains untouched.*

## Introduction

The entry rule is that if a stock's 15-minute return is unusually weak relative to the index's 15-minute return, the entry is made and the stock is held for a maximum of 15 minutes. This does not require that the stock's level have actually declined. A stop can be set to limit a loss or exit the position before it recovers, while a target can be used to take a profit or to cap any potential additional gain. By keeping entries and position sizes the same, this study can isolate the selected exit rules in this experiment.

The main question is whether adding the selected stop and target changes mean net daily return relative to a fixed-time exit. Stop-only comparisons and downside-risk measures are secondary outcomes. Candle ambiguity and bootstrap reliability are supporting diagnostics, not claims for a profitable strategy. The combined exit would be favoured on mean return if the sample has a positive contrast, while time-only would be favoured if the sample has a negative contrast. These signs alone are not sufficient to create a trading recommendation or establish a population effect. The signal subtracts index returns with a coefficient of one, and the position is an unhedged long position in the stock. This coefficient is not an estimated beta; it is a design convention.

Minute candles will show a range of prices but they may not show which barrier was reached first. For purposes of this report, path sensitivity refers to the difference in a matched comparison of two policies between the declared open-low-high-close and open-high-low-close sequences. A conditional ranking is one that holds under both paths and the stated fill rules. It does not establish the historical tick sequence, all possible price paths, actual fills or broad robustness.

The topic is the same as M2 and M4 and the five-stock design is the same as well. The present revision is based on the review dated 17 October 2026 by Ikemefula Solomon Oriaku and Zione Esther Kamtambe (Oriaku and Kamtambe). The existing results are complemented by gross/net decomposition, economic scale, development sensitivity and tail-loss diagnostics. A separate point-by-point response explains changes made, existing evidence and recommendations that were not applied in the fixed experiment. The preparation of drafts and checks on organisation and code were assisted by AI (OpenAI).[1]

**Table 1. Feedback and review status**

| Feedback addressed | Response in this draft |
|---|---|
| Limit the core scope to be manageable. | Five fixed development stocks and three exits for one entry rule, extensions after completion of the matched experiment. |
| Make a study-specific contribution | Matched exit comparisons, gross/net decomposition, explicit ambiguity counts and reproducible accounting. These empirical results are conditional; the final test remains separate. |
| Update and use appropriate references | The focused framework integrates basic techniques and recent cost-aware intraday research and official calendar evidence. |
| Explain the reasoning, and connect the sections. | Brief explanations connect the literature to the matched experiment, differentiate risk measures and identify what is still to be tested. |
| Body text should be justified. | Justified body paragraphs, readable tables and consistently formatted citations are used throughout the report. |
| Peer review | Acknowledged and responded to in the accompanying response. The subject remains the same throughout and the final test is not used; restrictions, omissions and extensions not performed are clearly indicated. |

## Theoretical Framework

### Reversal and execution

Jegadeesh reports negative first-order serial correlation in monthly stock returns (881). This gives historical evidence of reversal but not over 15 minutes. Using standardized factor-residual scores, Avellaneda and Lee trade hedged portfolios (763, 769–70). We use a fixed coefficient of one for our stock-minus-index return, and the position is long only; it is not an estimated factor residual or a market-neutral trade. Thus, we test exit rules given this entry signal, but do not assume the signal has a profitable edge.

There are other intraday patterns besides reversal. Gao et al. show that the first 30 minutes' return is a predictor of the last 30 minutes' return in US ETFs (394). It is a different horizon and market, but a warning not to think that every intraday price move is fleeting. Sampath and Gopalaswamy document pronounced opening and closing patterns in volatility and trading activity (271) for India. This encourages comparison at the same clock time; it does not validate our threshold, holding period or exit levels.

However, Heston, Korajczyk and Sadka identify short-lag reversal signals in quote-based returns in the United States, while their decile strategies lose when trades cross the spread (9, 14). De Groot, Huij and Zhou retain estimated weekly reversal profits in their samples of stocks in the US and Europe (3) after changing their stock selection and turnover. Different horizons, markets and cost assumptions constrain direct transfer to this NSE experiment.

[1] AI assistance saved time on organisation of review comments, code inspection and comparison of report values with saved outputs. It is not an evidence source for the market. The report text was also subjected to language refinement (Humantonebot). The numerical results, citations and technical meaning require independent checking; neither tool establishes authorship, empirical validity or conformance to a detector threshold.

Such studies encourage the separation of gross returns and costs, rather than assuming that reversal is profitable.

Aleti, Bollerslev and Siggaard break training, validation and testing apart chronologically (7736–37), and simulate intraday trading of ETFs with given spread and borrowing assumptions (7739–40). This fixed long-stock rule is different from their models and exposure. The comparison encourages an examination of timing and expense, but does not provide evidence for the current signal.

**How stopping and targeting can have a positive or negative effect**

In their independent, identically distributed return model with a positive risky-asset premium, Kaminski and Lo demonstrate that the stop overlay does not increase expected return (242). Their mean-reverting AR(1) analysis indicates a first-order return penalty, but if momentum persists, the outcome may differ (243–44). They have a portfolio-switching rule and a frictionless theoretical environment, distinct from the trade-level barriers here (240). A negative result for this strategy does not prove their mechanism, nor does it prove that there is no exit rule that can improve any unprofitable entry rule.

Leung and Li's stop-constrained Ornstein–Uhlenbeck model is a model that connects the stop level to the optimal take-profit boundary (15). It does not derive the illustrative multipliers employed here. Lo and Remorov demonstrate that return and risk rankings can differ: some relaxed stops may yield a higher mean-variance certainty equivalent or less drawdown at the expense of raw returns relative to buy-and-hold (9–10). Aggarwal provides an Indian pairs-trading comparison with daily closing prices and no spread adjustment (55–57). These differences prevent direct claims of replication.

These studies provide reasons for both gains and losses from early exits. Thus we use the same entries and position sizes, to demonstrate what difference the exit rule makes.

**Observation and evaluation**

Corwin and Schultz estimate spreads with high–low ratios over one- and two-day intervals (719). Their estimator does not recover the quoted spread at a specific exit or the queue position of a limit order, but provides a potential daily cost proxy. We are thus still making cost and fill assumptions, which remain scenarios. It is not possible to determine from the price data that all touched targets would have filled.

A candle that eventually gets to both barriers can leave an exit of an active position unresolved, as Maier-Paape and Platen show (17). This known problem is measured in the chosen NSE sample – how frequently it does occur, how this impacts exit-policy comparisons, and which assumptions alter the conclusions.

Sullivan, Timmermann and White discuss how the ability to pick a surviving rule from numerous trials can create a false sense of success (1–3). In the present study, one signal is maintained and the declared execution sensitivities are recorded. The study does not apply their Reality Check. Acerbi and Tasche's definition of expected shortfall includes fractional probability mass at the cutoff (5–6, 9–10); Yamai and Yoshiba demonstrate that the precision of the estimation depends on the sample size and the shape of the tail (89–90, 95–96). An estimate of a sparse tail is not necessarily precise even if the arithmetic is done correctly.

This evidence does not establish that such an NSE signal or such barriers will be profitable. It specifies the separation that the experiment must make between the effect of the exit rule, assumptions for its execution and the uncertainty of the reported estimates.

# Methodology

## Conventions used for samples and data

The pilot utilizes RELIANCE, HDFCBANK, INFY, TCS and ITC with the supplied .NSEI series during January–March 2021. The five names are not a random or reconstructed historical liquidity sample, but rather a manageable case study fixed before the recorded strategy outcomes. The extended source calendar extends to 2021–2023; however, preparing later files does not include assessing later strategy outcomes. OHLC and activity measures are not able to establish historical quoted spreads, depth or position in the queue. The results apply only to the names, dates and assumptions analysed.

On the NSE calendar, 26 January, 11 March and 29 March are marked as first-quarter holidays on weekdays, which leaves 61 trading dates in the first quarter of the year (National Stock Exchange of India, "Trading Holidays for the Calendar Year 2021"). The Securities and Exchange Board of India (SEBI) have noted the 24 February suspension and the extended session (1–2). It also says NSE submitted that NIFTY computation was unavailable between 10:06 and 11:43 (8). These sources describe the event, but do not prove that our files are complete.

The pilot does not include the full day of 24th February and keeps this day as missing history. The next 20 sessions are also barred under the rule and eligibility can be restored from 26 March. This will intentionally dispose of all otherwise useable windows on 24 February. We fixed the restriction before computing P&L for pilots, but the trader would not have been able to know about the event beforehand. The exclusion does not give any indication of loss per outage day nor a minimum performance.

For now we have assumed 09:15 to 15:30 Asia/Kolkata time and are studying both the start- and end-labelled minutes, pending supplier confirmation. We do not select an interpretation for better returns. The parser recognizes explicit input dates in the form MM/DD/YYYY and DD-MM-YYYY, and rejects any other date format, invalid OHLC and non-minute timestamps. In-month timestamps have to be strictly increasing and duplicate keys are rejected. An out-of-month row is excluded from the result set only when there is an explicit documented exception, its timestamp is the same as a row already retained from the original monthly source and the OHLCV values are all identical. If there are unmatched or conflicting rows, the ingestion is stopped. Verified copies have their own exclusions count and all the rows in the source are counted in the audit count and also in the parsed-content digest. Raw files are not altered. Where documentation is not available, adjustment conventions, historical charges, tick sizes and source metadata remain unresolved.

The 5 stocks selected had 1,883 copied observations in their September 2023 cash files dated 22 August 2023. Every one of these corresponds to the same August time stamp and OHLCV values: RELIANCE, HDFCBANK and TCS – 377 copies each, INFY and ITC – 376 copies each. These copies appear in the loader, but as verified copies, are not changed and are excluded by the loader. No observations were available for this date (22 September 2023) in the selected cash files; August prices are not used to reconstruct this date. The rules for feature-history and outcome-identification as declared above remain valid for missing data. This is a finding related to the quality of the sources, not to the actual performance of the final period strategy.

## Signals and corresponding exits

We subtract the log return from the completed 15-minute interval for the index from the log return for the stock during the same interval at each decision time. We determine the z value of this difference based on the mean and sample standard deviation of differences for the same clock time in the twenty scheduled sessions that immediately precede the session that

contains it. The calculation is not valid if history is missing or if it has zero variance. Valid dates that are older than the window cannot be used to fill in missing dates in the window.

Under start labels, a 09:45 decision uses the log of the close labelled 09:44 / close labelled 09:29. All sixteen consecutive bars must be on both instruments. Then end labels are shifted back 1 minute and the same calculation is applied. Decisions are taken every 30 minutes after 30 minutes of the session begins. A z score of −2 or lower allows for a long position at the beginning price one minute later. The full 15-minute holding window must fit within the session.

Define P as the entry price, and let v be the sample standard deviation of the stock's own same-clock 15-minute log returns for those 20 sessions. The stop is $P \times \exp(-v)$ and the target is $P \times \exp(1.5v)$. These are fixed, non-trailing points that continue the design that has been submitted. The signal is based on relative weakness and the distances of the barriers are based on the volatility of the stock. The z-value is not a calibrated probability and the barriers are not estimates of the optimal exit level.

**Table 2. Matched exit policies**

| Policy | Exit rule |
|---|---|
| Time-only | Close at common scheduled expiry. |
| Stop-only | At that expiry unless closed at the stop. |
| Stop and target | Close on the first barrier triggered otherwise at the expiry. |

For every delay scenario, all policies share the same entries, allocations and expiry. Each entry uses $0.5 \times C0 / N$: INR 10,000 for five stocks and reference capital C0 of INR 100,000. The research positions are on a fractional share basis and have a maximum of one position per stock. Early exit proceeds are kept aside until the common expiry time. It is a design which compares exits on the same opportunities, without separately optimising each strategy, or having the ability to exit early to fund further trading.

### The pros and cons of a fixed design

A manageable case study is assured through the five stock universe, implementation and audit work. It is not a random sample nor a recorded order of historical liquidity nor can it reflect the entire exchange. Keeping the same names will prevent changing the sample based on the results.

Comparisons are made at the same clock time with 20 scheduled sessions of history representing approximately one trading month. This is NOT an empirically best lookback, but a short reference window. The threshold of -2 for z scores is not a calibrated tail probability, but an unusually low index-adjusted return given that window. One illustrative asymmetric pair in log-price space is between the 1v stop and 1.5v target. Not necessarily assumed to be optimal or to provide a realised reward-to-risk ratio. Reference cost and checks of 15 bp and 5/25 bp respectively are assumed but not validated historical rates.

These explanations are included post-development and serve to explain the current design. They do not show a previously undocumented selection process. No change of the parameters or selected stocks because of the findings.

### Execution and costs

The minute prices display the range traded, and not necessarily which barrier was reached first. Both stated price paths are compared, and the trades for which this uncertainty is relevant are counted.

The engine uses open-low-high-close and open-high-low-close paths in a uniform manner for all policies. When the active long position's candle opens between the barriers and touches both, then these paths create stop first and target first results. These are not recovered ticks, but possible sequences, nor are they universal drawdown bounds. Opening price dictates the exit at scheduled expiry before any subsequent candle tops or bottoms. Prior to the expiration, an opening gap through the stop will fill at the open with the specified adverse slippage. No price improvement and gap above the target fills at the target. A threshold touch is still an assumed fill.

The reference round trip cost is 15 bps of entry notional with 1/2 at entry and 1/2 at exit. We change one assumption at a time; total cost to 5 or 25 bp, extra stop slippage to 5 or 10 bp, or entry delay of two or five minutes. The same fixed charge cancels when matched policy returns are subtracted, but this does affect a policy's absolute return. The actual charges according to exit value are not required to cancel. These checks focus on selected execution assumptions and do not provide checks for all sources of spread, market-impact or fill risk.

**Estimands and accounting**

Each pilot day starts independently with C0. The main estimate is the average delta P&L of fixed-notional sessions over C0, for those dates for which all the required signals and all policy outcomes are measurable. If one of the required signals cannot be calculated, the entire day's primary comparison won't be available; this is not the same as a valid signal that results in no trade. This estimate is also conditional on a complete case, and on the basis of this, the independent-session comparison was also performed. It outlines the evaluable dates, but not performance on every date and the viability of cash between dates.

These rules are implemented in the notebook after the frozen pilot and before validation or final-test strategy outcomes are examined. The funded diagnostic will record an abstention if pre-entry signal information is unavailable. If a bar doesn't provide an entry/exit value or held position value, then the outcome is unknown, not that there was no trade. Diagnostics make it possible to differentiate between abstentions, valid no-trade days, incomplete primary coverage, unresolved outcomes and funding failures. Funded diagnostics do not substitute an alternative sample for the primary complete-case sample.

A separate feasibility investigation is done for funded accounting. C0 is always the starting reference capital for each call to development, validation and final-test. In that diagnostic, cash and realised P&L are carried over days, notionals stay fixed and costs are charged on fills. There is no reinvestment of the gains and borrowing is not allowed. The matched funded comparison is stopped when the first economic funding failure occurs. Record keeps date, account affected, available cash and known equity, required funding with fees. An exit-fee cash deficit also puts an end to the comparison. The previous funded path is descriptive, since its endpoint depends on outcomes and it is not a substitute for the main fixed-calendar comparison. A failure cannot be deleted or followed by a cash reset within that funded segment.

If one does not know the market price when confirming an equity account, a cash amount is known, and the account has no open positions, then missing market prices need not interrupt its equity record. Unresolved post entry P&L does break the funded record until resolved. Starting new days later with independent money can't make up for an unknown funded cash balance. A gap is not a place where a new cash reset is allowed in the same funded segment. If there is a separately reported restart with prior P&L not reconciled, the restart must be labeled as a new independent segment with explicit initial conditions and cannot be connected with the wealth path before the restart.

Equity includes cash and open positions marked to market. Drawdown measures for saved minute-close and known-flat boundary marks, and for completed daily closes in the

notebook, are reported separately. They both begin with the opening capital of the recorded segment. Daily-close equity is the identified closing cash of completed, flat funded sessions. A partial mark from an interrupted day is not a daily close. If no identified day is completed in a segment the statistics for the daily close are not available, and separate segments are never joined. Fixed-capital normalisation is the P&L divided by C0, not a return on changing opening wealth. The mean of the worst 2.5% of daily losses, with fractional cut-off weighting, is ES97.5. The independent-session accounts adhere to common evaluable dates across policies; each policy's ES is individually calculated on those common dates before subtraction. Funded drawdown uses its own sample and endpoint – ES is descriptive. Independent-session means that each day the capital is reset, not that dates are statistically independent. Neither frequency of valuation identifies losses hidden within the minute.

**Constraints of the execution experiment**

The round-trip fee is applied to the entry notional and by construction it cancels from the matched exit-policy contrasts. Its scenarios show sensitivity of absolute returns, not relative execution costs. Stop-specific slippage is varied separately. The real cost of a stop and a limit target need not be the same: a stop can cross a spread and a limit target can fail to fill. These differences are not identified by the supplied candles. Gap-through-stop execution at the open impacts both stop-based policies. Capping a target-gap fill at the target forgoes favourable price improvement, while filling every target touch might be overly optimistic. These assumptions do not establish that the combined-policy estimate is uniformly conservative.

Fill probability, spread series and capacity estimates are not invented. Without dated records of constituent weights, a combined historical index weight is also not reported. The unhedged position continues to have market exposure and inclusion in the index can lead to a somewhat mechanical relative-return signal. Different questions would be tested by a random-entry benchmark, estimated-beta signal, larger stock universe, alternative lookbacks and barrier grid. They remain explicitly unperformed extensions and not hidden modifications of this fixed comparison.

**Rules for Evaluation and Completion Plan**

The primary comparison is the combined-minus-time mean net daily return at reference costs, entry delay and stop-first ordering. The hypothesis of no difference is the null hypothesis. Stop-minus-time and combined-minus-stop are secondary comparisons. Combined-minus-stop is an indicator of how much the target adds to the selected stop. There is no target-only policy or estimate of the total stop–target interaction.

The 60/20/20 split is based on the full-period calendar and not the dates that satisfy the complete-case requirement. In this instance, we have n sessions planned; development goes up to session floor(0.60n), validation up to session floor(0.80n), and the final test uses the remaining sessions. The recorded boundaries are 1 January 2021–17 October 2022, 18 October 2022–25 May 2023 and 26 May 2023–29 December 2023, based on 739 normal sessions. The calendar covers the holiday amendment (National Stock Exchange of India, "Trading Holidays for the Calendar Year 2022"; National Stock Exchange of India, "Trading Holidays for the Calendar Year 2023"; NSE Clearing Limited) made in June 2023. Validation was finished on 18 October 2026 and final-test strategy outcomes remain unevaluated. Later exclusions cannot move these boundaries. Features could be initialised from previous observations. Validation may lead to bugs being found or metadata corrections based on independent evidence, but not to undocumented searches for more favourable thresholds, stocks or exclusions. Amendments will be noted prior to final testing and the original pilot and any test results already observed will be retained.

A nominal 95% paired moving-block percentile interval, based on 2000 repetitions and seed 17838, is used for the notebook. Day blocks of ten and twenty are sensitivities; day blocks of five are primary blocks. Whole paired daily observations are sampled within each contiguous eligible calendar segment. The blocks are sampled uniformly with replacement and concatenated to the original segment length except for the sampled block at the end, which is truncated. No original date is removed from the point estimate. The interval is withheld if any segment is shorter than its block length, or if $\sum_j \lfloor L_j/b \rfloor < 10$. In this expression, $L_j$ is the number of eligible sessions in segment $j$, and $b$ is the block length in sessions. This support rule does not guarantee independence, coverage or power – it is merely a reporting threshold. It is a variant of fixed-block resampling that does not recover missing outcomes and is based on the observed gaps (Künsch). The development update has synthetic reliability and interval-width diagnostics along with market results. Group review of validation and final freezing of the protocol remain pending. The lengths of the blocks should not be chosen for statistical significance.

All of the currently nominal 95% bootstrap intervals are exploratory. Interval not including zero indicates the direction according to the assumed resampling model, but not a statistically reliable effect. Synthetic diagnostics indicate under coverage. Additional repetitions or selection of a favourable block length do not resolve that limitation. Such restriction shall be for development, validation and final outputs of the present procedure. Sensitivity tests illustrate the results' sensitivity to the assumptions made. There is a trade-off between lower sampled risk and lower returns.

This exploratory status must be maintained with the protocol review or a separately specified inferential amendment must be documented and evaluated before final outcomes are accessed. That assessment must consider the following: dependence and missing-date patterns, interval coverage and width, and simulation uncertainty. If there is no supporting evidence for defensibility, then descriptive effect estimates and exploratory intervals will be retained in the final report. The software checks or the saving of a protocol freeze does not create statistical reliability.

Tail-risk estimates may not be sufficiently precise with sample size expansion. There are 148 scheduled sessions in the fixed final partition, which is a maximum of 3.70 days of scheduled ES97.5 tail weight without exclusions. This is not an actual final-test sample; it is a calendar calculation. Final ES will be descriptive unless backed up by documented precision assessment. The notebook shows the actual number of sessions, the tail dates and tail leave-one-day-out sensitivities for each stage performed. Independent tail events cannot be formed with more trades or bootstrap repetitions.

**Table 3. Completed work and remaining stages**

| Stage | Status / required evidence |
|---|---|
| Inherited pilot and development evidence | Completed historical records. All of the tables (5–10) and their original run identifiers are preserved. They are not new M6 market runs. |
| M6 report development | Explanatory links, source corrections and the dated working plan have been added. M4 feedback and the peer review received are dealt with and each decision is noted in the accompanying response. |
| M6 notebook checks. | Independent trade/cash/day and episode reconciliation, chronological boundary checks and runtime package recording are implemented. All 25 of the notebook's code cells were completed on 18 October, and all 111 tests and 14 market ledger reconciliations were successful. Peer-review revision includes separate aggregate diagnostics checked without rerunning the market experiment. These checks are detailed in Appendix B. |
| Resolution of sources and metadata. | Verified copied rows remain audited. Timestamp meaning and adjustment conventions/execution metadata are not determined if supplier evidence is not available. |
| Validation | All 148 scheduled sessions are evaluable for both timestamp interpretations in all seven settings in the run completed on 18 October 2026. Results and manifests are kept separate from development. |
| Final evaluation | Final strategy outcomes for the 26 May - 29 December 2023 period are not evaluated. The interim M6 notebook does not allow for the final evaluation or freezing of a protocol. |

## Precision and interpretation

There is no claim of a validated minimal detectable effect. By way of example, if the development start-label interval, with width 0.5180 bp, is assumed to be a symmetric normal 95% interval, the approximate standard error is 0.1321 bp. For a two-sided 5% test with 80% power, the effect required would be approximately 2.80 times that standard error, or 0.3700 bp per session. If the variance, dependence and availability were not changed, the scaling by the square root of 360/148 yields approximately 0.5771 bp for 148 sessions. These are assumptions only and because of the described undercoverage, this is a planning indication only and not a power guarantee. It does not involve final outcomes.

Use of a greater number of bootstrap replications decreases the variation in the simulation but does not correct for an incorrect or inadequate resampling method. The 200-history diagnostic is evidence of failure under tested conditions, not calibration of a reliable 95% procedure. Stationary bootstrap and sign-flip methods would necessitate justified dependence, symmetry and missingness assumptions. Repeated trades on the same dates do not establish independent observations. No multiplicity-adjusted claims of significance are made - all current contrasts and execution sensitivities are descriptive/exploratory.

White's Reality Check and Hansen's superior predictive ability test are both tests for comparing a benchmark with a given family of alternatives and taking this comparison process into account (White 1097; Hansen 365). When a strategy is chosen from a set of candidate backtests, Bailey et al. evaluate overfitting. These methods are not just about how many stocks; they are about the set of rules in question. These tests are not reported as completed. The main contrast is constant, and the extra parameter and execution checks are descriptive, not a search for a winning rule.

The random sign-flip test does not necessarily apply to serially dependent daily differences. It assumes that the joint null distribution is transformation-invariant under the transformations used (Hemerik and Goeman, sec. 2.1). Zero mean alone does not establish that condition. If the signs are flipped independently each day, then this cannot be labelled an

exact test of the mean as it can change serial dependence, unless additional assumptions are made.

### Group working plan

The remaining group work for the M6 deadline of 3 November 2026 (at 00:00 GMT / 01:00 in Berlin) is provided in Table 4. The internal completion date is 2nd November which leaves time to review the files. These are not statements of completed activities, but are planned responsibilities and dates. Access to the final test is beyond the scope of this interim release until a validation review and documented protocol freeze occur.

**Table 4. Group working plan to the M6 deadline**

| Work and lead | Target | Completion evidence |
|---|---|---|
| Code and reconciliation: Uchenna | 18–22 Oct. | Review and reproduce the validation run carried out on 18 October, checking the manifests and the ledger checks and recorded limitations. |
| Data review: Israel | 18–23 Oct. | Examine source counts, exclusions, calendar boundaries and outstanding metadata. Leave both timestamp interpretations open where unresolved. |
| Analysis and references: Goodness | 23–27 Oct. | Review each table for consistency with saved results, provide a discussion of results and limitations, verify citations and add or remove from reference list. |
| Peer feedback: all members | 22–28 Oct. | Review the point-by-point response with the revisions and make note of group decisions for the other extensions. |
| Draft review: all members | 28–30 Oct. | Review the research question, method, results and conclusion for consistency; verify other tasks and interpretation constraints. |
| Reproduction and file checks: all members | 31 Oct.–1 Nov. | Reproduce the submitted outputs, check the figures and the units, inspect the PDF and check that the supporting files open correctly. |
| Group sign-off and upload | 2 Nov. | Complete the group review and provide the PDF separately from the supporting ZIP prior to the course deadline. |

Following the M6 draft, the group will take a further look at the protocol from 4-6 November and record any amendment prior to final access. A separate final-release notebook is to be provided 7-9 November, subject to that review and in line with the instructor's final-project timetable. The current M6 notebook keeps on blocking final execution and freezing. They are not official new deadlines, but internal targets.

While timestamp conventions remain unconfirmed, we retain both conditional interpretations. In cases of sparse coverage, inference may not be available. Where ambiguous episodes are absent, we report this absence, but do not modify barriers to manufacture dual hits.

## Results

The pilot and development results provide the initial evidence for M6. The new validation results are presented in a separate subsection. The 3 stages are not pooled, but separated by dates and run identifiers.

### Frozen pilot: implementation and coverage

The original M4 pilot added streamed ingestion, strict parser, past only signals and session accounting to M3. Independent calculations verified the reported signals, executions and daily

accounts in that version which passed 99 checks. These are historical checks for Tables 5-7, not current notebook test count or proof of historical fills. Later, self-contained notebooks and verification records are detailed in the development update and Appendix A. The code can be found in the link: https://github.com/I-am-Uchenna/nse-intraday-risk-controls.

The frozen pilot sample includes 137,036 rows from eighteen monthly files that are held by the instructor in the NSE Cash Stocks Data and NSE Index Data. These selected rows were successful in the pilot parser checks and missing intervals were left in. This is the count of the original Q1 sample. The later source audit commands its own manifest and is not able to certify instruments and files beyond the study inputs selected.

**Table 5. Session / Episode coverage of reference execution settings**

| Measure | Start labels | End labels |
|---|---|---|
| Scheduled Q1 sessions | 61 | 61 |
| Initial history-only sessions | 20 | 20 |
| Nonstandard session excluded | 1 | 1 |
| Subsequent sessions with an incomplete prior history | 20 | 20 |
| Common evaluable sessions | 20 | 20 |
| Valid zero-trade sessions: | 4 | 7 |
| Unique eligible trade episodes. | 33 | 24 |
| Unresolved episodes | 0 | 0 |
| Ambiguous episodes of active dual-barrier (ADB) | 0 | 0 |

From Frozen M4 replay - results/m4_pilot/summary.json Twenty evaluable dates are available: 1–23 February (seventeen sessions), 26 March, 30 March and 31 March. Label interpretations change signals and entries, counts are unique episodes and not repeated policy/path accounts.

## Conditional comparisons

**Table 6. Reference mean daily P&L / reference capital, in basis points**

| Policy and/or paired contrast | Start labels | End labels |
|---|---|---|
| Time-only | -2.914 | -0.886 |
| Stop-only | -2.734 | -1.359 |
| Stop and target | -3.476 | -1.957 |
| Combined minus time | -0.562 | -1.072 |
| Stop minus time | +0.180 | -0.473 |
| Combined minus stop | -0.742 | -0.598 |

Means and paired contrasts produced from scenario_means.csv and paired_contrasts.csv. The default settings are 15 bp round trip, no additional stop slippage and a 1 minute entry delay. Each day starts independently with capital of INR 100,000 with each basis point being INR 10. Both paths agree because no active episode encounters both barriers in one candle.

**Table 7. Sensitivity of paired mean daily differences (bps of reference capital)**

| Setting | Start: combined-time | Start: stop-time | End: combined-time | End: stop-time |
|---|---|---|---|---|
| Reference | -0.562 | +0.180 | -1.072 | -0.473 |
| Common cost 5 bp | -0.562 | +0.180 | -1.072 | -0.473 |
| Common cost 25 bp | -0.562 | +0.180 | -1.072 | -0.473 |
| Stop slippage +5 bp | -0.962 | -0.220 | -1.297 | -0.698 |
| Stop slippage +10 bp | -1.362 | -0.620 | -1.522 | -0.923 |
| Entry delay 2 min | -0.071 | +0.476 | -1.204 | -0.641 |
| Entry delay 5 min | -0.997 | +0.027 | -1.466 | -0.667 |

Data from paired_contrasts.csv and sensitivity_comparisons.csv. Each setting have exactly the same twenty evaluable dates. Entries need not be preserved in case of delay and/or label changes. Common fixed costs are eliminated in the comparisons and absolute returns continue to vary. For instance, end-label time-only has an average of +0.314 bp per day at a 5 bp cost, whereas stop-only and combined are still negative.

At the reference cost, all three policies lose, on average. Combined exits do not perform as well as time-only in all the stated settings. The small stop-only advantage of the start-label becomes negative with the addition of 5 bp stop slippage. The result has been obtained by the development pilot. In the case of 20 equally weighted daily outcomes, empirical ES97.5 would be the same as the worst observed loss. Hence, the pilot does not report any ES, drawdown, Sharpe ratio or inferential interval.

## Expanded development results

### Completed scope and data audit

The extended development period is 1 January 2021 to 17 October 2022, and uses the same five stocks, entry rule and three exits. It contains seven execution settings under two different timestamp interpretations; these fourteen settings are related, not independent experiments. Preparation includes 216 selected symbol-month entries for 6 series, retaining 1,669,406 rows in the period 2021-2023. This does not guarantee complete minute coverage or coverage of all instruments delivered. The unreconciled June 2023 supplement is still not included. Source preparation is not a strategy evaluation. The exact run, code and protocol identities are provided in Appendix A.

The source audit records the verified 1,883 August copies and also the missing cash information for 22 September 2023. The hosted development run succeeded and all 24 code cells ran without any errors. The M4 revision has passed 104 software tests and a second cached-pilot run reproduces all 84 historical policy/scenario means to within a maximum of 0.000000000001 bp. These computation checks are helpful for having the ability to replicate results under the assumptions. They are not evidence of historical execution quality.

### Independent-session estimates

**Table 8. Comparison of development at reference execution**

| Measure | Start labels | End labels |
|---|---|---|
| Scheduled development sessions | 443 | 443 |
| Common evaluable independent sessions | 360 | 381 |
| Mean daily P&L / C0 (bp) - time only | -2.3362 | -2.3863 |
| Stop-only mean daily P&L / C0 (bp) | -2.5925 | -2.4935 |
| Combined mean daily P&L / C0 (bp) | -2.5687 | -2.5137 |
| Combined minus time (bp). | -0.2325 | -0.1273 |
| Primary five-day exploratory 95% interval (bp) | [-0.4410, 0.0770] | [-0.3269, 0.1299] |
| Time-only ES97.5 daily loss (bp). | 17.4085 | 15.3083 |
| Daily loss (bp) for the stop-only group (ES97.5) | 14.8495 | 14.2981 |
| Daily ES97.5 loss (bp) for combined exits | 14.4075 | 13.2774 |
| The tail mass (days) of the ES97.5. | 9.000 | 9.525 |
| Unique ambiguous / eligible episodes. | 0 / 637 | 0 / 662 |

Data source: This run's policy_means.csv, paired_effects.csv, expected_shortfall.csv, intervals.csv and exit_ambiguity.csv. Reference execution: 15 bp round-trip cost, one-minute delay and no extra stop slippage and O-L-H-C ordering. Means and ES: common independent-session days, missing outcomes are not zero returns. The intervals are still exploratory. Any interval that includes 0 (including when an endpoint is 0) is inconclusive. ES is descriptive. These development observations are not final tests and are not intended as a substitute for the original pilot.

The development combined-minus-time estimates are negative but imprecise. Both the five- and ten-day-block exploratory intervals contain zero under both timestamp interpretations, and modest gains are possible (along with losses under the resampling model). Twenty-day intervals are withheld since the first eligible segment is only seventeen days. Lower observed ES97.5 does not establish overall superiority. The lack of a dual-barrier reference episode means that it is not evidence about performance during ambiguous candles if the two path results are equal. The new diagnostic tables below numerically reveal the development sensitivities and fragility of the tail estimates.

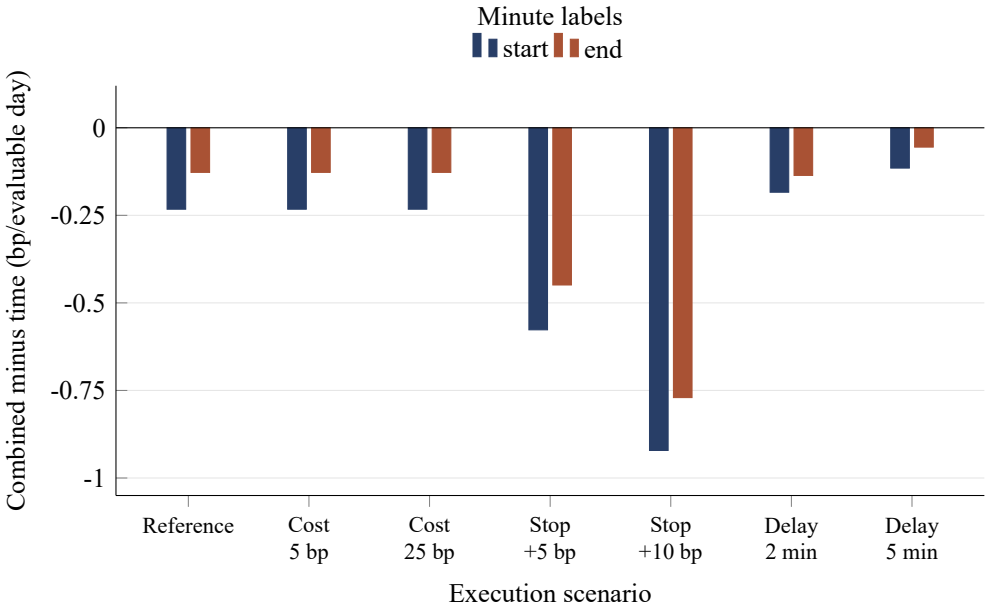



Figure 1. Combined minus time mean effects for the declared execution scenarios. The following comparisons are aligned on reference and stressed dates within each interpretation of the timestamps. The Y axis is basis points (bps) of reference capital per evaluable day. The figures plotted are point estimates, not confidence intervals, and these are development comparisons.

### Funded equity and drawdown

**Table 9. Funded drawdown for development on reference execution**

| Minute labels / policy | Maximum drawdown (%) at minute and boundary marks | Daily-close maximum drawdown (%) | Identified daily closes | The most recent date with an identified daily close. |
|---|---|---|---|---|
| Start: Time-only | 9.4282 | 9.3359 | 443 | 2022-10-17 |
| Start: Stop-only | 10.4371 | 10.3714 | 443 | 2022-10-17 |
| Start: Combined | 10.1413 | 10.1191 | 443 | 2022-10-17 |
| End: Time-only | 9.9072 | 9.8252 | 443 | 2022-10-17 |
| End: Stop-only | 10.1650 | 10.0845 | 443 | 2022-10-17 |
| End: Combined | 10.0296 | 10.0116 | 443 | 2022-10-17 |

Source: funded_drawdown.csv, O-L-H-C reference accounts. The six accounts shown each contain one identified segment within its timestamp interpretation in the time range, 1 January 2021 to 17 October 2022. None of the funded accounts are early ending; there are 443 identified daily closes for each account. The beginning of each segment begins at its recorded C0. The maximum drawdown at minute marks includes all the saved marks and does not include the downsampling plotting. Daily-close drawdown is based on finished, identified sessions and an independent daily-close peak. There is no requirement for the independent-session day count in the funded sample, where there are known abstentions. A failure or unidentified position may end the minute record during a day, after the last daily close. Equity between multiple segments cannot be joined; initial conditions for each segment need to be identified.

Under both interpretations of the timestamp and for both valuation frequencies, both of the stop-based policies exhibit a higher funded maximum drawdown than time-only. Reducing independent-session ES is not necessarily a reduction in funded drawdown. Additional identified abstention days and trades permitted when other candidate information is unavail-

able are included in the funded diagnostic. It has a different sample and cash path compared to the primary complete-case comparison.

### Bootstrap reliability diagnostics

The added diagnostics use the specified resampler on Gaussian paired daily differences with mean 0 and stationary standard deviation 1 bp: independent observations and AR(1) coefficients 0.3 and 0.7. Each process evolves over all 443 of the scheduled development dates before the actual 360 day or 381 day eligibility mask is applied. These diagnostics evaluate repeated-sample interval coverage and width for these specified processes, in addition to arithmetic tests. The experiment is carried out using 200 simulated samples per design, the settings of the experiment, the seed and the output are reported in bootstrap_reliability.csv. This is a zero mean coverage experiment, no power is claimed on a trading advantage.

**Table 10. Synthetic bootstrap diagnostics**

| Mask and AR(1) coefficient | Coverage %, MCSE pp | Average width bp, MCSE bp | Average / Oracle width (ratio) |
|---|---|---|---|
| Start, AR(1) 0.0 | 90.5 (2.07) | 0.1993 (0.0010) | 0.965 |
| End, AR(1) 0.0 | 94.5 (1.61) | 0.1964 (0.0010) | 0.978 |
| Start, AR(1) 0.3 | 91.0 (2.02) | 0.2530 (0.0015) | 0.902 |
| End, AR(1) 0.3 | 89.0 (2.21) | 0.2485 (0.0014) | 0.910 |
| Start, AR(1) 0.7 | 78.5 (2.90) | 0.3368 (0.0025) | 0.696 |
| End, AR(1) 0.7 | 81.5 (2.75) | 0.3334 (0.0024) | 0.705 |

Coverage ranges from 78.5% to 94.5%. These are respectively 78.5% for start labels and 81.5% for end labels with an AR(1) coefficient of 0.7. The nominal 95% procedure may thus have actual coverage less than the nominal level for these simple stationary processes. Monte Carlo standard errors (not confidence limits) are given in parentheses in the table. Each row has 200 intervals from independent histories, with 2,000 resamples per interval and five-day blocks, for each design. The exact calculated Gaussian width, based on the known covariance and retained dates, is also compared with the mean width. The process seed is 20261002; the bootstrap seed is 17838. The histories shared between the two masks are simulated and are not independent experiments per row. This diagnostic was established after the original development run but before validation or final-test evaluation. This is not a 'pre-registration' that applies to past events. The caching arithmetic can re-create the original interval function within 1e-11 bp for three paired probe vectors for each mask. These checks are given for the observed masks, and for a fixed bootstrap seed. They do not model missingness as a function of some outcome, heavy tails or regime change, nor provide coverage for the true NSE process or recover missing dates. Unsupported market intervals are still not available. We report undercoverage as a limitation, and don't select a "block length" based on the effect. Group review of validation and the final freeze are still to be done.

### M6 validation - fixed design on later sessions

The validation run was performed with the same five stocks, entry rule, barriers and execution settings and completed on 18 October 2026. It evaluates 18 October 2022-25 May 2023. In all settings, all 148 scheduled sessions are evaluable under both timestamp interpretations. The 216 selected symbol-month entries were again read from the instructor's Drive archives, retaining 1,669,406 rows after the same 1,883 verified duplicate exclusions. Their parsed-

data fingerprint matches the development record. Final-period source preparation remains separate from strategy evaluation: no final-test outcomes were evaluated.

**Table 11. Validation returns and risk at reference execution**

| Labels / exit | Mean (bp/day) | ES97.5 (bp) | Minute / boundary MDD (%) | Daily-close MDD (%) |
|---|---|---|---|---|
| Start / Time | -2.0100 | 13.3647 | 3.0080 | 2.9851 |
| Start / Stop | -2.2864 | 11.6941 | 3.4170 | 3.3891 |
| Start / Combined | -2.6985 | 11.8599 | 4.0344 | 4.0140 |
| End / Time | -2.0510 | 13.9353 | 3.0476 | 3.0354 |
| End / Stop | -2.4511 | 11.8667 | 3.6397 | 3.6276 |
| End / Combined. | -2.4365 | 12.0332 | 3.6181 | 3.6060 |

Table 11 assumes 15 bps round-trip costs, a one-minute entry delay and O-L-H-C ordering. The normalisation of daily P&L and ES is based on INR 100,000. Each ES gets only 3.70 days of effective tail weight; the worst three daily losses get full weight and the fourth gets 0.70. There are also 148 daily closes identified on the funded accounts with no economic funding failures. Despite the fact that the date counts now match, the accounting is still different. Drawdowns measure saved marks, not unseen intraminute extremes.

The mean difference for combined-minus-time is $-0.6885$ bp per day (start labels) and $-0.3855$ bp (end labels). The nominal five-day-block 95% intervals are $[-1.7037, -0.0809]$ and $[-1.2170, 0.1695]$ bp, respectively. The first does not include zero, but both are exploratory due to the reported coverage limitation. Ten-day-block intervals are $[-1.7444, -0.0607]$ and $[-1.2630, 0.1873]$ bp. The twenty-day intervals are withheld: 148 sessions deliver seven support blocks, less than the stated minimum of ten. This is not a notebook cell failure, but a reporting rule.

The descriptive ES of both stop-based exits is lower than that of time-only exits, while the mean returns are lower and the funded drawdowns are larger for both. The mean under start labels is decreased by 0.4120 bp and the mean under end labels is increased by 0.0146 bp with the addition of a target to the stop. Such mixed signs do not support a general benefit from the target. The combined-exit ES leave-one-day-out range is 11.2186-11.8669 bp under start labels and 11.4182-12.0384 bp under end labels. These are sensitivity ranges, not confidence intervals.

**Table 12. Validation execution sensitivity**

| One changed assumption | Start labels (bp/day) | End labels (bp/day) |
|---|---|---|
| Reference | -0.6885 | -0.3855 |
| 5 bp round-trip cost | -0.6885 | -0.3855 |
| 25 bp round-trip cost | -0.6885 | -0.3855 |
| Extra slip on stops: 5 bp | -1.0838 | -0.7335 |
| Extra stop slippage: 10 bp | -1.4790 | -1.0815 |
| Entry delay: 2 minutes | -0.2592 | -0.2919 |
| Entry delay: 5 minutes | -0.4418 | -0.1920 |

Table 12 reports combined minus time in bp of reference capital per day, using O-L-H-C ordering and the same 148 dates for each row. All seven settings have a negative contrast. Equal fixed-notional fees cancel in the paired comparison by construction; this is not evidence of general cost robustness. An extra 10 bp stop penalty degrades the contrast by

0.7905 bp for start labels and 0.6959 bp for end labels. Delayed entries affect the comparison and barriers but do not reverse the sign of the comparison.

Validation has one reference ambiguous episode under end labels: 1 of 260 (0.3846%), which is not the case for development. There are no ambiguous episodes among 266 start-labelled episodes at reference execution. For end labels, the combined mean is $-2.4365$ bp under O-L-H-C and $-2.4093$ bp under O-H-L-C, a difference of 0.0272 bp per day. The contrast for the combined-minus-time shifts from $-0.3855$ to $-0.3583$ bp, both of which are negative. One stop exit is changed to a target exit. These episode counts are for reference execution only - start-labelled two- and five-minute delays also yield path differences, both of approximately 0.0239 bp/day. This is only partial evidence of the impact of ordering of the candles on the declared sample and is not sufficient to draw general conclusions about path robustness or the actual sequence of past executions.

Run 20261018T084332Z-ghd27dnw records these outputs. No Python errors are stored in the code cells after Drive was connected, all 25 code cells completed, and the saved code cells match the archived 18 October source. The later peer-review revision changes export bookkeeping and introduces aggregate diagnostics (checked separately). All 14 of the ledger reconciliations were within the INR $10^{-7}$ tolerance, with the biggest discrepancy being under INR $2.73 \times 10^{-11}$. These policy, path and accounting records do not constitute independent observations, as they are repeated. The checks establish internal arithmetic consistency within their stated scope, but not real fills, reliable interval coverage or profitability.

**Peer-review diagnostics from stored aggregates**

These diagnostics are based on the development and validation exports recorded. They add no entries, tweak no parameter and check no final-period result. The notebook creates the same four tables, using its current-stage aggregates, and the supporting script reproduces the tables from the recorded CSVs.

**Table 13. Time-only gross return and cost decomposition**

| Stage / label | Gross bp/day | Fee bp/day | Net bp/day | Gross bp/trade |
|---|---|---|---|---|
| Development / start | 0.3179 | 2.6542 | -2.3362 | 1.7968 |
| Development / end | 0.2200 | 2.6063 | -2.3863 | 1.2660 |
| Validation / start | 0.6860 | 2.6959 | -2.0100 | 3.8167 |
| Validation / end | 0.5842 | 2.6351 | -2.0510 | 3.3252 |

Daily basis points use reference capital of INR 100,000. Entry notional is used for trade basis points. A 10% share of the reference capital is allocated to each episode; gross daily return is the same as the net daily return plus 15 times 0.10 times the number of episodes divided by evaluable days. All four time-only gross means are positive, but their 1.2660-3.8167 bp per-trade scale is less than the assumed 15 bp fee. This is not a comparison with random entry. The combined-minus-time differences are about INR -2.32 and -1.27 per day for the start and end labels, respectively, in development and INR -6.88 and -3.86 per day in validation. These are conditional averages at the specified allocation.

**Table 14. Calendar, estimation and funded-account samples**

| Stage / label | Sched. | Complete | Unavailable | Funded | Episodes |
|---|---|---|---|---|---|
| Development / start | 443 | 360 | 83 | 443 | 637 |
| Development / end | 443 | 381 | 62 | 443 | 662 |
| Validation / start | 148 | 148 | 0 | 148 | 266 |
| Validation / end | 148 | 148 | 0 | 148 | 260 |

Scheduled dates, primary complete-case dates and identified funded dates are different counts. The 83 and 62 unavailable development dates comprise one explicit exclusion plus 82 and 61 incomplete candidates. A funded account may remain identified while holding cash on a date that has no eligible entry. Its 443 dates do not imply 443 complete-case daily contrasts. The 21-day gap between the timestamp conventions can be carried through twenty-session histories; it is not known to be attributable to any single outage. The difference in the count does not exist in validation.

**Table 15. Development combined-minus-time sensitivity**

| Setting | Start bp/day | Start days | End bp/day | End days |
|---|---|---|---|---|
| Reference | -0.2325 | 360 | -0.1273 | 381 |
| Cost: 5 bp | -0.2325 | 360 | -0.1273 | 381 |
| Cost: 25 bp | -0.2325 | 360 | -0.1273 | 381 |
| Stop slippage: 5 bp | -0.5769 | 360 | -0.4489 | 381 |
| Stop slippage: 10 bp | -0.9214 | 360 | -0.7704 | 381 |
| Entry delay: 2 min | -0.1840 | 360 | -0.1360 | 381 |
| Entry delay: 5 min | -0.1151 | 360 | -0.0552 | 381 |

Each execution comparison uses the intersection of dates available under that scenario and reference execution. These intersections still contain 360 and 381 days in development and 148 days in validation. The combined-minus-time contrasts are negative in every listed setting. All three policy contrasts and changes from reference, as well as aligned counts for both stages, are included in the supporting CSV. None is considered an independent strategy-selection success.

**Table 16. Expected-shortfall sensitivity to removing one day**

| Stage / label | Policy | Tail days | ES97.5 bp | Leave-one-day-out bp |
|---|---|---|---|---|
| Dev. / start | Time | 9.0000 | 17.4085 | 14.9946–17.4221 |
| Dev. / start | Stop | 9.0000 | 14.8495 | 14.3377–14.8540 |
| Dev. / start | Combined | 9.0000 | 14.4075 | 13.7943–14.4125 |
| Dev. / end | Time | 9.5250 | 15.3083 | 14.1607–15.3189 |
| Dev. / end. | Stop | 9.5250 | 14.2981 | 13.2745–14.3044 |
| Dev. / end | Combined | 9.5250 | 13.2774 | 12.1298–13.2850 |
| Val. / start | Time | 3.7000 | 13.3647 | 11.8750–13.3811 |
| Val. / start | Stop | 3.7000 | 11.6941 | 11.1532–11.6987 |
| Val. / start | Combined | 3.7000 | 11.8599 | 11.2186–11.8669 |
| Val. / end | Time | 3.7000 | 13.9353 | 12.0263–13.9584 |
| Val. / end | Stop | 3.7000 | 11.8667 | 11.4177–11.8717 |
| Val. / end | Combined | 3.7000 | 12.0332 | 11.4182–12.0384 |

The last column is the minimum and maximum ES97.5 with each observed day deleted in turn and the tail mass recomputed. It is not a confidence interval, but rather a sensitivity range. This table shows descriptive losses in bp of reference capital under reference execution and O-L-H-C ordering. The 3.7 tail-day mass in validation calls for particularly cautious interpretation. The remaining final period is not used to increase the sample.

These deletion ranges do not indicate whether the policy ranking survives when one shared worst date is deleted. That matched-date ranking check has not been performed.

## Discussion

Lower expected shortfall is not the same as a lower drawdown. We report both, and discuss their samples and accounting, before judging whether an exit policy improves risk.

Evidence from the extended development does not support a mean-return benefit for combined stop-loss and profit target exits. Combined exits result in lower mean returns per evaluable day than time-only exits under start- and end-labelled minutes, respectively, by 0.2325 and 0.1273 basis points. This is based on 443 scheduled sessions with 360 and 381 complete case sessions. Lower mean returns also are achieved in all declared cost, slippage and delay settings when combined exits are made. The nominal intervals are exploratory due to synthetic diagnostics showing undercoverage. In their mean-reverting model (243–44), Kaminski and Lo give reasons why a stop may miss a recovery. We find that our result is consistent with that worry, but do not identify their mechanism, as their portfolio overlay isn't the same as ours, and we haven't estimated their return model.

Understanding of the policy comparison is supported by development execution sensitivities. For 10 bp stop slippage, the combined-minus-time difference gets worse by 0.6889 bp and 0.6430 bp for start and end labels respectively. With a 5-minute delay the difference becomes less negative by 0.1174 and 0.0722 bp, but its sign does not change. Common fixed charges cancel from matched contrasts in the stated fee model, but still affect absolute returns. Entries and barriers can change with delay and timestamp changes, so these scenarios do not isolate causal effects with trades held constant. The previous 20-day pilot is a summary of the implementation. Its small stop-only advantage does not survive an added-slippage check.

Expected shortfall and funded drawdown are different concepts. Expected shortfall summarises the worst daily losses on common complete-case dates. Funded drawdown follows capital over time and can accumulate a persistent negative drift. Its ranking therefore need not provide independent evidence about volatility or tail protection. Different accounting samples are used for the two measures in development; validation uses the same dates but different accounting definitions. Detrending could be a secondary diagnostic but would strip out some of the money an investor would actually lose and cannot replace the funded drawdown reported. Lo and Remorov also find that return and risk rankings may not be the same (9–10).

The candle ambiguity problem was previously established by Maier-Paape and Platen (17). No eligible reference episodes in the development sample on common evaluable dates touch both barriers within one active candle: the observed counts are 0/637 and 0/662. Both declared paths thus agree on this sample. These counts are neither population limits nor evidence of robustness to ambiguous candles. The study provides a reproducible comparison of the submitted exits with explicit limits on execution, missingness and inference. Synthetic both-hit cases test implementation, rather than the historical significance of ambiguity. The policy outcomes, paired contrasts, common coverage, ambiguity counts and exit reasons will be carried over to the final report. The validation results above add one ambiguous end-labelled episode; final evaluation is pending.

Stressful or other unusual dates may be excluded from complete-case analysis. With a twenty-scheduled-session history, a signal window on later dates may be invalidated by

a missing required observation; it need not cost only one evaluation date. A one-minute timestamp shift changes the raw labels picked up at window edges and can propagate through those histories. This is not an attribution of all missing days, but a mechanism for different counts. Scheduled, complete-case and funded dates are now shown separately in the report. When the required observations themselves are missing, it is not possible to perform a direct excluded-day volatility or signal-frequency comparison. Using the most recent available dates would alter the definition of the history and is not silently substituted.

**Table 17. Obstacles and responses**

| Obstacle | Response and remaining limit |
|---|---|
| Unconfirmed interval labels or time zone | Compare both interpretations for diagnostics, seek confirmation from the source and hold back final claims which are dependent on the timing. |
| Deliveries overlapping or conflicting with other deliveries. | Maintain the source identity, only removing copies which are known to be identical and out of month. Any rows that are unmatched or have conflicts will prevent the ingestion. |
| Missing intervals and sessions not held at the usual time | Do not backfill any gaps or forward fill prices; retain exclusions and missing history, and do not silently select complete trades. |
| Unknown spreads, impact and actual fills | Use declared cost/fill scenarios; do not claim executable returns or capacity. |
| Limited days for independent assessments | Describe estimates, present intervals as exploratory until coverage and tail precision have been evaluated. |
| Funded-record interruption | Keep the first failure or unknown outcome and its prefix; never concatenate unreconciled segments. |

This record can be used by researchers to make a decision whether or not to further investigate the selected exit controls. The same audit method can be used to audit broker or platform backtests, where the results of bracket-orders rely on a candle sequence that isn't disclosed. Other markets may have researchers who can modify the approach to their respective calendars, instruments and costs. There is no evidence of performance in those markets which is provided by the NSE results themselves.

## Conclusion

The combined-minus-time point estimates are negative in the development and validation phases. Their uncertainty remains exploratory and the study does not establish a population return advantage or the absence of any possible benefit. Lower observed expected shortfall accompanies lower mean returns and larger funded drawdown. Positive gross time-only returns are not high enough to offset the assumed reference cost. Path ordering matters in one end-labelled validation reference episode but does not reverse the return ranking; the sparse ambiguity evidence cannot establish general robustness. The contribution is a reproducible comparison of fixed exits with explicit accounting, execution and inference limits. The topic, entry rule and final-test boundaries are unchanged. The final test has not been evaluated.

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## Appendix A. Reproducibility

The runnable M6 notebook can be found at https://colab.research.google.com/github/I-am-Uchenna/nse-intraday-risk-controls/blob/main/NSE_Intraday_M6.ipynb. The M6_README.md file in the repository provides links to the clean source, recorded validation outputs and verification evidence.

The commit of the historical pilot implementation is commit 494ca1299696ce0968a1eb39fa14018628f60a68, which can be found on https://github.com/I-am-Uchenna/nse-intraday-risk-controls/tree/494ca1299696ce0968a1eb39fa14018628f60a68. It's configured with the JSON values from commit ab23883; its manifests are support for Tables 5-7. The notebook used for the completed development run is commit 8b40cf6e4170a157095ecfc28d0579cf1299c99a, notebook SHA-256 0c4bfa9a55f2de2a31604b2dd2add76abee0344208ae124cbd8149e4d5a03eaf. The development manifest contains its "run" identity, its "parsed-data" identity, its calendar, and its execution assumptions and inference settings. Records of both are maintained.

The current M6 notebook contains 26 code cells, 24 of which are unchanged original cells, one of which is the revised export cell and one is the new aggregate-diagnostics cell. The recorded validation run on 18 October completed 25 code cells, using Python 3.13.16, NumPy 2.1.3, pandas 2.2.3 and Matplotlib 3.10.0. The development run used Linux with Python 3.13.15, NumPy 2.1.3 and pandas 2.2.3, and Matplotlib was used for figures. The pilot used Python 3.12.14. The new diagnostics have separate aggregate-only checks. Shared outputs exclude source prices, private ledgers and course documents. Synthetic checks do not require the original instructor archives, while market replay does.

To get started, open the NSE_Intraday_M6.ipynb notebook in Colab, enter the paths of the instructor's folders and run the cells in order in a hosted CPU runtime. The notebook selects sources and stores them in a disk-based database, while the strategy reads one session at a time. Choose pilot for Q1, development for the development partition or validation for the next declared stage. This interim M6 notebook blocks final-test execution and protocol freezing. A later final release will need validation review and a matching versioned freeze before final-test access. Every ingestion attempt is given a new path for the database; partial earlier ingestion attempts are saved.

The original twenty day comparisons are kept in both results/m4_pilot/summary.json and CSVs. The development run is recorded and it exports separate policy means, paired effects, coverage, status counts, ambiguity, funded drawdown, expected shortfall, tail dates, intervals and source/protocol manifests. The synthetic diagnostic CSV file includes data about coverage, Monte Carlo error, width, oracle width and unavailable counts; bootstrap_reliability_audit.json keeps settings, seeds and precise eligibility masks. The two output generations cannot replace each other. Numbers are stored at full precision and rounded to display. Instructor prices and private ledgers with details are not included in the common aggregate-output ZIP.

NSE_Intraday_M6.ipynb contains the implementation of M6. For each timestamp interpretation and execution setting, the inherited experiment calls run_one_scenario and run_research. The methods and analysis can be seen in the notebook. The M6 additions check boundaries and independently reconcile trade, cash, day and episode records before accepting the exported comparisons. There were 19 synthetic method scenarios and five focused M6 tests, all of which passed. These checks establish software behaviour on their test cases; they do not verify a market result or guarantee statistical coverage.

**Table 18. Notebook guide**

| Notebook section | Functions and purpose |
|---|---|
| 1-3. Run instructions, Research design, Calendar and chronological split | RUN_STAGE; CASH_FOLDER; INDEX_FOLDER; SAVE_FOLDER; CFG. Set the input/output locations, declared rules and chronological evaluation stage. |
| Load instructor data | discover_sources; vendor_rows; build_database. Find specific inputs, check the validity of the inputs from a vendor, and feed these into a disk-based SQLite database. |
| 4. Signal and sizing | window_feature; plan_session; replay_session; run_research. Develop past-only signals, configure matched entries, utilize the three exit policies and use session/funded accounts. |
| 5. Uncertainty and tail risk (paired) | expected_shortfall; paired_block_interval; bootstrap_reliability_diagnostic. Calculate descriptive tail loss, exploratory paired intervals and synthetic coverage/width diagnostics. |
| 6. Run the experiment | run_one_scenario; RUNS. Perform two timestamp interpretations in seven scenarios and save the results of each scenario. This is driving the market-analysis. |
| 7-12. Sample, returns, funded risk, intervals, sensitivity and conclusions | ALL_DAILY, MEANS, FUNDED_RISK, INTERVALS, ALIGNMENT. Summarize exclusions, returns, drawdown and uncertainty; compare execution assumptions and interpret limits. |
| Peer-review diagnostics | Four aggregate tables: gross/net returns, sample accounting, aligned execution sensitivity and ES leave-one-day-out sensitivity. No additional market replay. |
| 13. Save and verify | Export aggregate tables, plots, provenance and reconciliation checks. In this interim release we have disabled protocol freezing and final-test access. |

The expanded development aggregates and original run manifests are located at https://github.com/I-am-Uchenna/nse-intraday-risk-controls/tree/main/results/development_20261001. The recorded outputs and their checksums are kept in this dated folder. The publication of it does not constitute a new market run.

### Recorded run and revised source

The original code and protocol are retained for the completed development run: 20261001T221531Z-th2hk303. The exploratory interpretation was made explicit in the 3 October notebook revision (commit 8d710c9e9fb1d6e1bb16dcdbcf08c6306911dddf). That revision did not change market data, trading rules or the numerical findings stored. The recorded M6 validation run has its own code identity, runtime record, source and protocol manifests and reconciliation results. Its protocol SHA-256 is 4f7c68c2570f32fc433588777a7f25ff46c3e7926aac4af62519a7a6666322ec. Its ordered code-cell payload SHA-256 is faf5c73927b69af815f16b607c8c2567ca25f477d270116cf9fabcbc0488f047; the checksum definition and exact downloaded-notebook checksum are in the verification record. The recorded M6 validation outputs are in results/validation_20261018 and the stage comparison is in results/m6_comparison_20261018. It does not inherit the previous run's verification status.

## Appendix B. Computational checks

The new M6 checks are distinct from the historical checks listed below. Cell research-027 tests permitted past history, rejection of later dates and rejection of a non-contiguous evaluation slice. Cell research-031 defines reconcile_ledgers; the m6-ledger-reconciliation cell independently recomputes trade P&L, daily totals, cash changes and matching across the six policy/path arms, and exports ledger_reconciliation.csv. In cell research-047, runtime

information is recorded. Final execution is blocked in cells research-002 and research-033; protocol freezing is blocked in research-002 and research-047. Market reconciliation passed for all 14 validation scenarios and is recorded separately from synthetic checks. This reconciliation performs arithmetic checks on exported account-days. It does not establish the correctness of fill prices or fee assumptions, recover missing outcomes, or independently verify funded cash continuity and drawdown. The software suite passed 111 tests with no failures or errors on 18 October. The added diagnostic cell was executed against both saved stages and matched all four exported tables. The complete revised software suite passed 115 tests with no failures or errors, including four new diagnostic tests. The peer-review additions have a separate verification record in the supporting package, and neither software check evaluates final-period market outcomes.

In the original pilot, the number of reference signals matched with the independent calculations was 57, the number of policy/path executions was 2,394 and the number of evaluable daily accounts was 1,680. The biggest daily P&L difference was less than INR 0.00000001. Repeated accounts do not constitute independent market data. The historical replay validated source counts and SHA-256 digests. It had a 384 MiB process limit, with preparation using 55.65 MiB and replay 39.70 MiB of total committed memory for all child processes (excluding the supervisor and operating-system cache). The preceding measurements are not of the later Colab notebook. Before the recorded development run, the then current notebook executed all 24 code cells and repeated the export in a separate check in the local cached-pilot. Its 84 means matched the historic pilot within 2.57e-13 bp, and had a maximum aggregate child-process committed memory of 332.66 MiB, with a 768 MiB limit. This is local verification memory and not hosted full-study memory. The hosted development run was then used to read the original Drive archives and run all 14 scenarios. Neither run assessed the validation or final-test strategy results.